

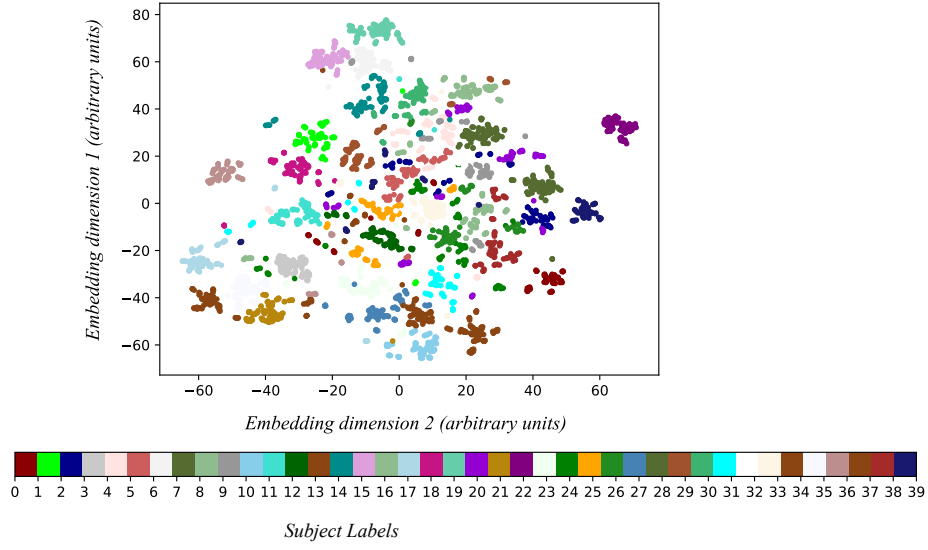


Figure 2. t -SNE of SPD covariance matrices using Riemannian distance indicates that the SPD matrices from different subjects lie in different neighborhoods of the manifold. This is due to shift in EMG signals owing to the combined effect of various anatomical, physiological, and circumstantial factors. Embedding is for NINAPRO: DATABASE 2-EXERCISE 1. Embedding is colored according to subjects. Each of the 40 subjects performed 102 trials (17 gestures, each repeated 6 times).

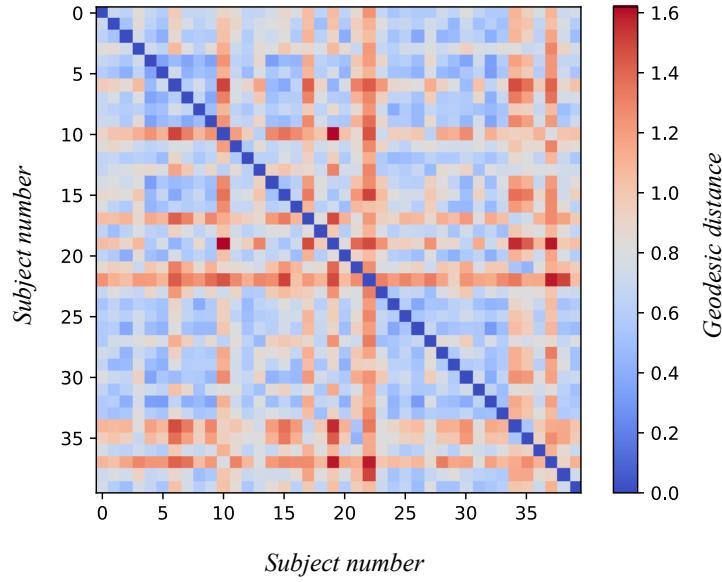


Figure 3. Riemannian geodesic distance between the centroids of SPD covariance matrices of 40 subjects in NINAPRO: DATABASE 2-EXERCISE 1. Geodesic distance quantifies the differences in EMG signals between subjects due to the combined effect of various physiological and anatomical factors. The centroid of a given subject is calculated as the *Log-Cholesky* average of SPD covariance matrices of all 102 trials (17 gestures, each repeated 6 times). X and Y axes are numbered according to subjects.

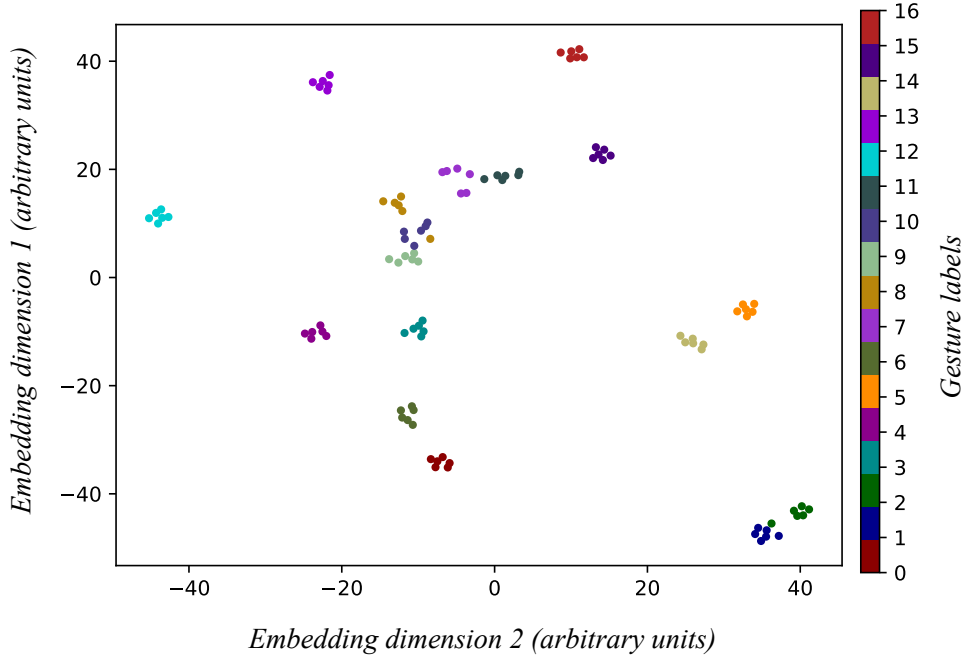


Figure 4. *t*-SNE of SPD matrices of *subject 0* from NINAPRO: DATABASE 2-EXERCISE 1 shows that different gestures within a subject have contrasting spatial patterns. We can classify these distinct gestures with supervised algorithms such as minimum distance to mean (MDM) and support vector machine (SVM) or unsupervised algorithms such as *k*-medoids clustering using Riemannian distance. Classification accuracy using the above methods is presented in appendix C.2 for all 40 subjects. Embedding is colored according to gestures. The subject performed 17 gestures with each gesture repeated six times.

The gestures are: 0: thumb up, 1: extension of index and middle - flexion of the others, 2: flexion of ring and little finger - extension of the others, 3: thumb opposing base of little finger, 4: abduction of all fingers, 5: fingers flexed together in fist, 6: pointing index, 7: adduction of extended fingers, 8: wrist supination (axis: middle finger), 9: wrist pronation (axis: middle finger), 10: wrist supination (axis: little finger), 11: wrist pronation (axis: little finger), 12: wrist flexion, 13: wrist extension, 14: wrist radial deviation, 15: wrist ulnar deviation, and 16: wrist extension with closed hand.

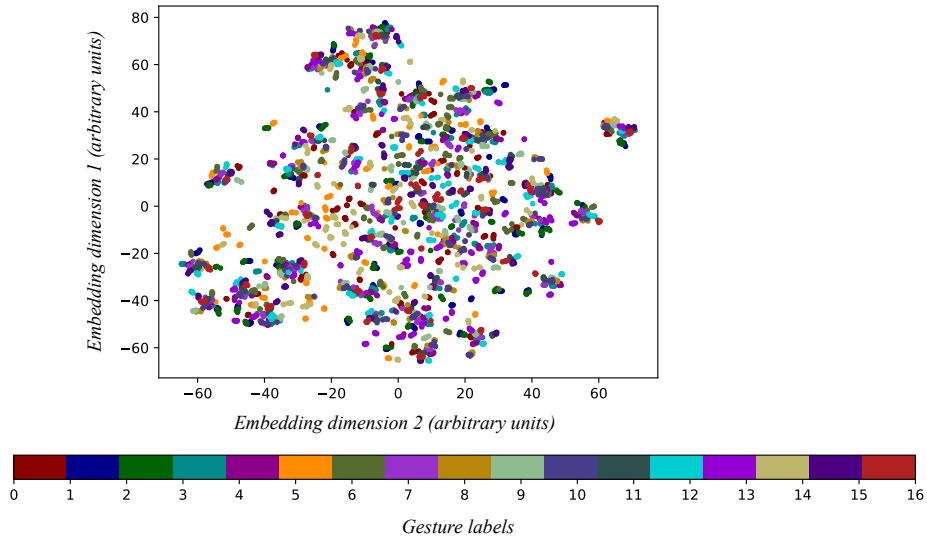


Figure 5. *t*-SNE of SPD covariance matrices using Riemannian distance without accounting for intersubject differences reveals that the SPD matrices for the same gesture from different subjects do not cluster together. This is the same embedding in figure 2 colored according to gestures (instead of subjects). Each of the 40 subjects performed 102 trials (17 gestures, each repeated 6 times). The gestures are: 0: thumb up, 1: extension of index and middle - flexion of the others, 2: flexion of ring and little finger - extension of the others, 3: thumb opposing base of little finger, 4: abduction of all fingers, 5: fingers flexed together in fist, 6: pointing index, 7: adduction of extended fingers, 8: wrist supination (axis: middle finger), 9: wrist pronation (axis: middle finger), 10: wrist supination (axis: little finger), 11: wrist pronation (axis: little finger), 12: wrist flexion, 13: wrist extension, 14: wrist radial deviation, 15: wrist ulnar deviation, and 16: wrist extension with closed hand.

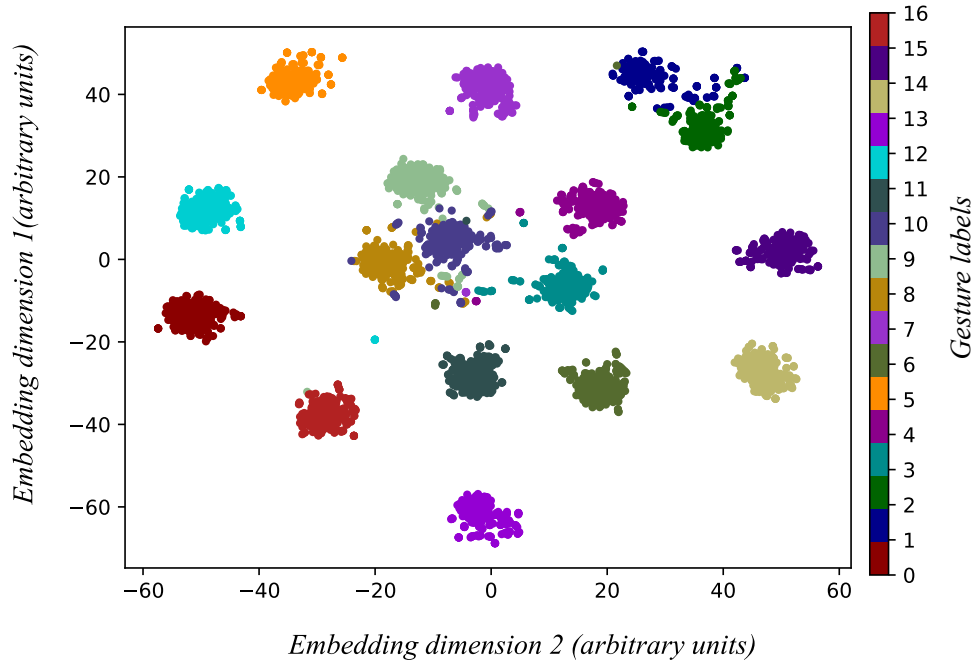


Figure 6. *t*-SNE of SPD covariance matrices using Riemannian distance after aligning the gestures using *parallel transport* as described in appendix B and illustrated in figure 7. Data is aligned assuming ground truth labels for visualization purposes. Otherwise, alignment accuracy is bounded by the accuracy of unsupervised classification. This demonstrates how data from different individuals in figure 5 can be aligned according to gestures so that the same gestures from different individuals are transported to the same neighborhood in the manifold space. Embedding is colored according to gestures. Each of the 40 subjects performed 102 trials (17 gestures, each repeated 6 times).

The gestures are: 0: thumb up, 1: extension of index and middle - flexion of the others, 2: flexion of ring and little finger - extension of the others, 3: thumb opposing base of little finger, 4: abduction of all fingers, 5: fingers flexed together in fist, 6: pointing index, 7: adduction of extended fingers, 8: wrist supination (axis: middle finger), 9: wrist pronation (axis: middle finger), 10: wrist supination (axis: little finger), 11: wrist pronation (axis: little finger), 12: wrist flexion, 13: wrist extension, 14: wrist radial deviation, 15: wrist ulnar deviation, and 16: wrist extension with closed hand.

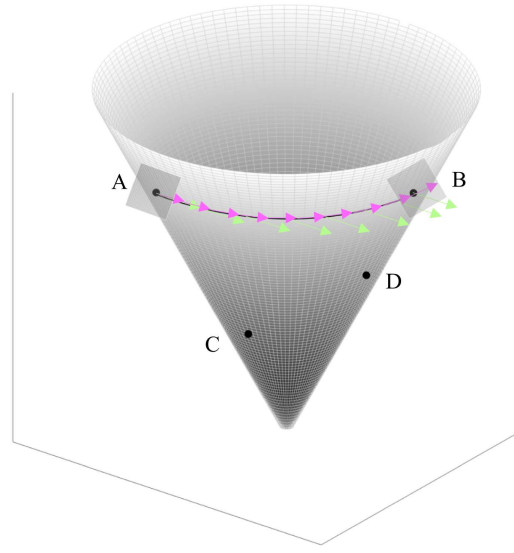


Figure 7. Euclidean transport is inappropriate in the manifold space. Point A is an SPD matrix that represents the EMG signal covariance for a given gesture, say a pinch, for a given subject. Point B is another SPD matrix representing the reference location for that gesture on the manifold: say, a particular reference subject's centroid for the pinch gesture. The length of the curve joining A and B is the geodesic distance between the two points. Points C and D , like A and B are SPD matrices that represent different gestures; for instance, the reference locations for a power grasp and a wrist flexion. *Green* arrows illustrate the Euclidean transport of a tangent vector from A to B . *Pink* arrows represent the *parallel* transport of a tangent vector from A to B . Parallel transport rotates the vector along the path from A to B so that the vector in the tangent plane of A remains in the tangent plane of B when transported. It can be seen that the Euclidean transport of a vector in the tangent plane of A results in vectors that are no longer in the tangent plane along the path to B . The Euclidean transport of vectors therefore cannot appropriately traverse a geodesic. The *square patches* at A and B represent the tangent planes.

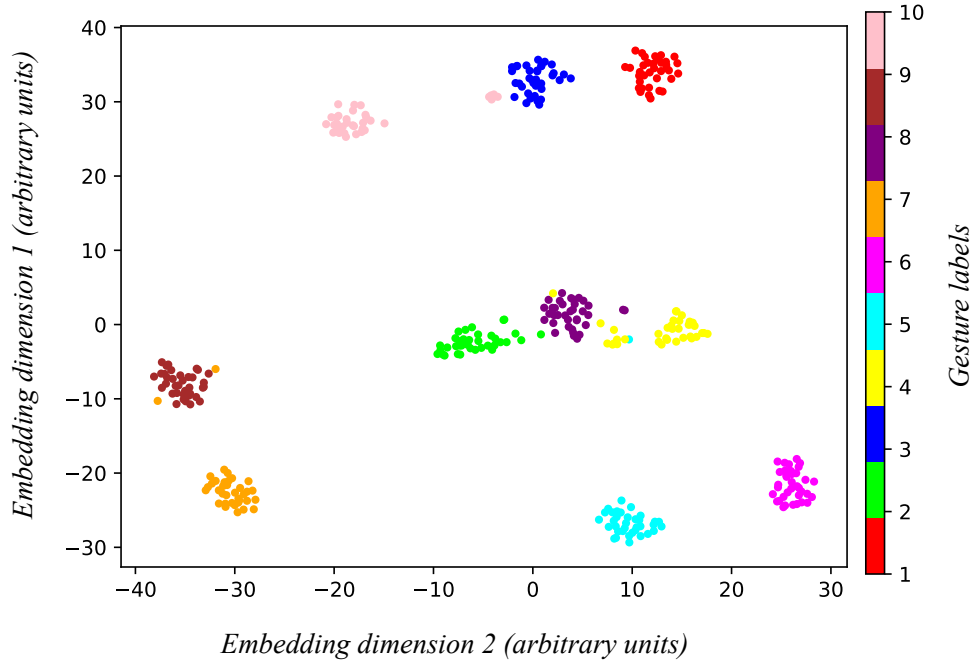


Figure 8. t -SNE of SPD matrices of *subject 0* from UCD-MYOVERSE-HAND-0 shows that different gestures within a subject have contrasting spatial patterns. We can classify these distinct gestures with supervised algorithms such as minimum distance to mean (MDM) and support vector machine (SVM) or unsupervised algorithms such as k -medoids clustering using Riemannian distance. Average classification accuracy across subjects using the above methods is presented in table 1. Embedding is colored according to gestures. The subject performed 10 gestures with each gesture repeated 36 times.

Ten gestures are: 1 - Down, 2 - Index finger pinch, 3 - Left, 4 - Middle finger pinch, 5 - Index point, 6 - Power grasp, 7 - Right, 8 - Two finger pinch, 9 - Up, 10 - Splay.